Transactional Data

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Outline

1. Transactional Data
   - Key Characteristics
   - Representation Formats
   - Applications
   - Challenges
2. Market Basket Analysis
   - Frequent Item Generation
   - Apriori Algorithm

Transactional Data

Introduction

1. Transactional data captures specific business transactions
2. Examples include retail sales, banking transactions, and online purchases
3. Often used in data mining, machine learning, and business intelligence

Transactional Data

Introduction

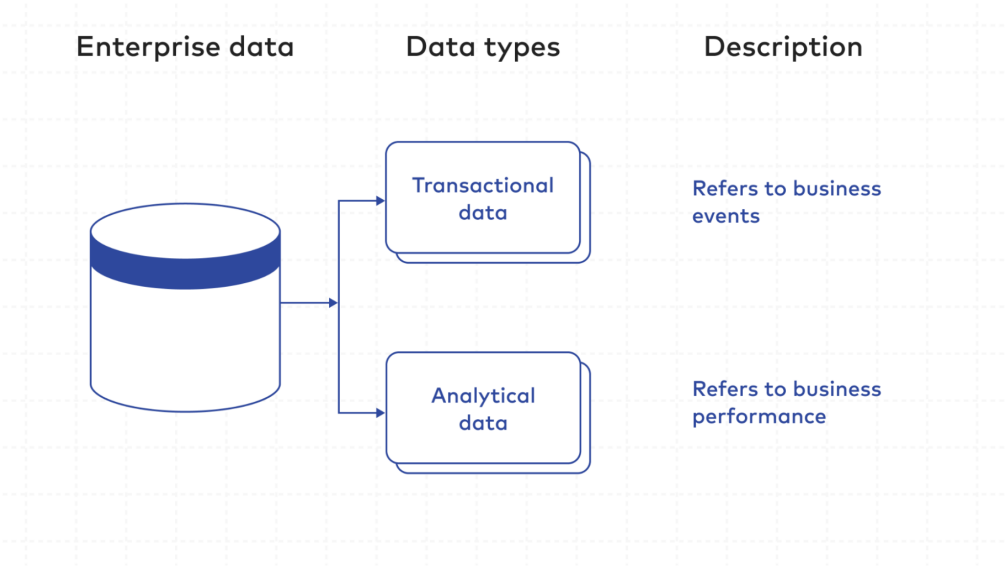


Figure: Transactional Data

Transactional Data

Key Characteristics

1. Time-Stamped: Each transaction is recorded with a timestamp
2. Structured Format: Organized into tables or records
3. Voluminous: Large datasets due to frequent transactions
4. Immutable: Original transaction records are not overwritten; instead, new records are added to reflect changes

Transactional Data

Key Characteristics

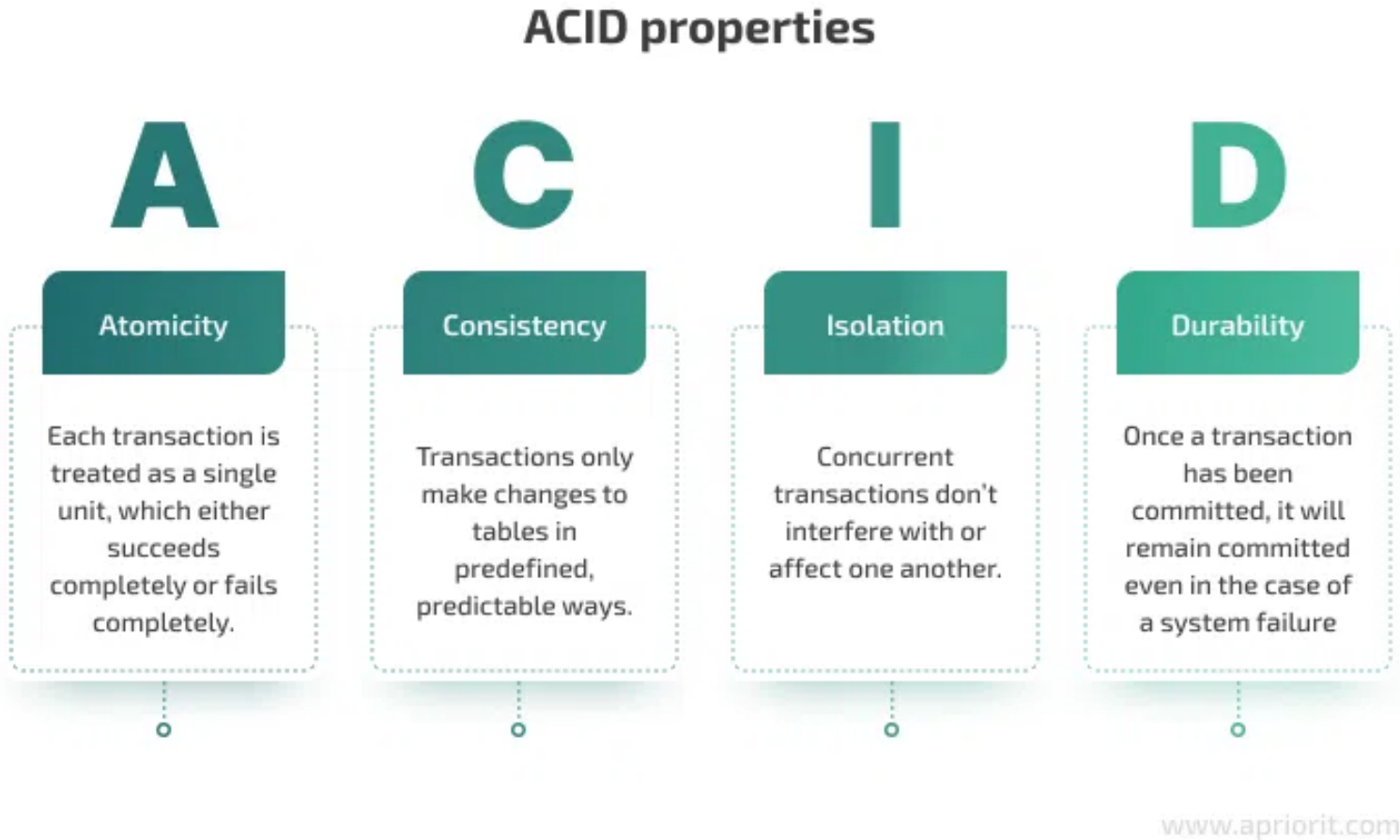


Figure: Key Characteristics of Transactional Data

Transactional Data

Key Characteristics

Atomicity

1. Ensures transactions are **all-or-nothing**
2. Example: In banking, either funds are fully transferred, or no transfer occurs

Consistency

1. Ensures data remains in a valid state after a transaction
2. Example: Total balance across linked accounts should remain consistent

Transactional Data

Key Characteristics

Isolation

1. Concurrent transactions should not interfere with each other
2. Example: Booking systems ensuring simultaneous ticket purchases don't overlap

Durability

1. Ensures committed transactions are permanently saved
2. Example: Data is safely stored even in case of system failure

Transactional Data

Representation Formats: Tabular Format

1. **Structure:** Rows for transactions, columns for features

ID	Item 1	Item 2	Item 3	Total	Date
1001	Apple	Banana	NULL	$5.00	2025-03-01
1002	Milk	Bread	Butter	$8.50	2025-03-02

2. Advantages: Simple, easily queried via SQL
3. Limitations: Less efficient for sparse data

Transactional Data

Representation Formats: Itemset Format

1. Structure: Transactions as sets of items
2. Example: {Milk, Bread, Butter}
3. Advantages: Ideal for market basket analysis
4. Limitations: Lacks transaction details like date, cost, etc.

Transactional Data

Representation Formats: Sequence Format

1. Structure: Transactions ordered by time
2. Example: T1 $\rightarrow$ {Milk, Bread}, T2 $\rightarrow$ {Apple}
3. Advantages: Useful for customer journey analysis
4. Limitations: Difficult to analyze out-of-order patterns; orders in a set might be unordered but still may mean the same thing from a business perspective

Transactional Data

Representation Formats: Graph-Based Format

1. Structure: Nodes as items, edges as relationships

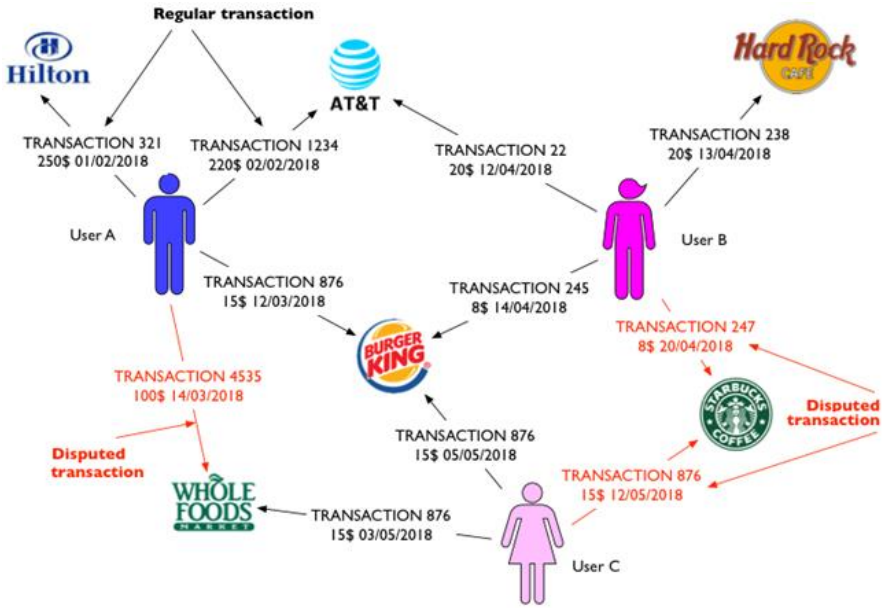


Figure: Graph-Based Format

2. Advantages: Captures complex item interactions
3. Limitations: Graph construction can be computationally expensive

Transactional Data

Representation Formats: Sparse Matrix Format

1. Structure: Rows as transactions, columns as items (binary values) in dense matrix are represented using $\{(row, column, value)\}$ format in its sparse representation

$$A_{dense} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A_{sparse} = \{(1, 1, 1),\ (2, 3, 1)\}$$

2. Advantages: Efficient for sparse data
3. Limitations: Inefficient for dense data

Transactional Data

Representation Formats: JSON, Key-Value Format

1. Structure: Flexible, hierarchical data structure

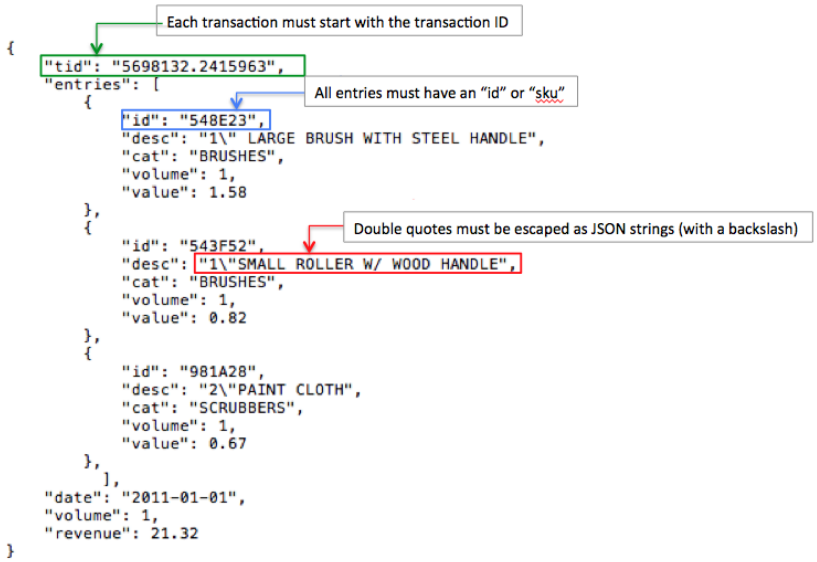


Figure: JSON-Based Format

2. Advantages: Flexible for modern web applications
3. Limitations: Requires parsing for traditional relational database systems; one must parse (break down / transform) the hierarchy into tables

Transactional Data

Representation Formats: Event Log Format

1. **Structure:** Logs with timestamps

Event ID	Customer ID	Event Type	Timestamp
E001	1234	Add to Cart	2025-03-01 10:05
E002	1234	Checkout	2025-03-01 10:10

2. Advantages: Suitable for behavioral analysis
3. Limitations: Can become large and require specialized tools

Transactional Data

Applications

1. Market Basket Analysis: Identifying item purchase patterns
2. Fraud Detection: Identifying suspicious activities
3. Customer Segmentation: Understanding buying behavior

Transactional Data

Challenges: Volume and Velocity

Large-scale transactional data generated at high speeds can overwhelm traditional systems

Mitigation Strategies:

1. Implement distributed storage systems (e.g., Hadoop, Spark)
2. Use data compression techniques to reduce storage overhead
3. Employ incremental data processing for real-time updates

Transactional Data

Challenges: Data Quality Issues

Description: Incomplete, inconsistent, or erroneous data can impact analysis

Mitigation Strategies:

1. Implement data cleaning pipelines
2. Use automated validation rules to detect anomalies
3. Apply imputation techniques to handle missing data

Transactional Data

Challenges: Data Sparsity

Description: Sparse matrices with numerous zero entries make computations inefficient

Mitigation Strategies:

1. Use specialized data structures like compressed sparse row (CSR)
2. Apply dimensionality reduction techniques (e.g., PCA, t-SNE)
3. Filter out infrequent items to improve data density

Transactional Data

Challenges: Data Privacy and Security

Description: Transactional data often contains sensitive information

Mitigation Strategies:

1. Use encryption techniques to secure data
2. Implement access controls and data masking
3. Employ differential privacy methods to protect individual records

Transactional Data

Challenges: Temporal Dependencies

Description: Transactions may have strong time-based correlations

Mitigation Strategies:

1. Use time-series models (e.g., ARIMA, LSTM) for trend prediction
2. Apply sequence mining algorithms for pattern detection
3. Implement windowing techniques for better temporal aggregation

Transactional Data

Challenges: Integration of Heterogeneous Data Sources

Description: Combining data from different sources can introduce inconsistencies

Mitigation Strategies:

1. Use ETL (Extract, Transform, Load) pipelines for data consolidation
2. Implement schema matching and entity resolution techniques
3. Employ knowledge graphs for improved data integration

Market Basket Analysis

Introduction

1. Market Basket Analysis (MBA) is a data mining technique used to uncover itemset (collection of one or more items that appear together) patterns in transactions
2. It identifies relationships between products in retail data
3. Applications include recommendation systems, inventory management, and marketing strategies

Market Basket Analysis

Introduction

1. MBA relies on identifying frequent itemsets in transaction data
2. Uses metrics like **support**, **confidence**, and **lift** to determine patterns

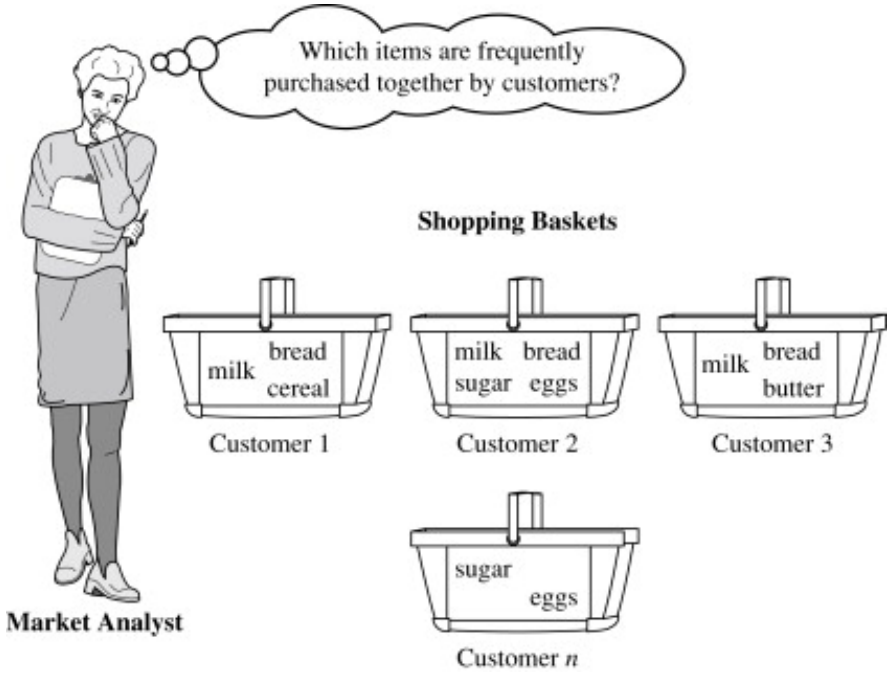


Figure: Market Basket Analysis

3. Example: Customers who buy *bread* often buy *butter* as well

Market Basket Analysis

Key Terminologies

1. Itemset: set of items purchased together
2. Support
   - A key measure used to evaluate how frequently an itemset appears in the dataset
   - Determines which itemsets are frequent enough to be considered useful for generating association rules
   - Mathematically, The support of an itemset X is defined as:

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

   - Ensures that the discovered patterns are statistically significant

Market Basket Analysis

Key Terminologies

Example: Consider the following transactions:

1. Transaction 1: {Milk, Bread}
2. Transaction 2: {Milk, Bread, Butter}
3. Transaction 3: {Bread, Butter}
4. Transaction 4: {Milk, Butter}

For the itemset $X = \{Milk, Bread\}$,

$$\text{Support}(\{Milk, Bread\}) = \frac{2}{4} = 0.5$$

Interpretation: The itemset {Milk, Bread} appears in 50% of all transactions.

Market Basket Analysis

Key Terminologies

Confidence

1. Measures the likelihood that a particular item (or set of items) is purchased, given that another item (or set of items) has already been bought
2. The confidence of the rule $X \rightarrow Y$ is given by:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

Where:

- $\text{Support}(X \cup Y)$ = Proportion of transactions containing both X and Y
- $\text{Support}(X)$ = Proportion of transactions containing X

3. A higher confidence value suggests a stronger association rule

Market Basket Analysis

Key Terminologies

Confidence: Example

Consider the following transactions:

1. Transaction 1: {Milk, Bread}
2. Transaction 2: {Milk, Bread, Butter}
3. Transaction 3: {Bread, Butter}
4. Transaction 4: {Milk, Butter}

Step 1: Compute Support Values

$$\text{Support}(\{Milk, Bread\}) = \frac{2}{4} = 0.5$$

$$\text{Support}(\{Milk, Bread, Butter\}) = \frac{1}{4} = 0.25$$

Market Basket Analysis

Key Terminologies

Confidence: Example

Step 2: Compute Confidence

$$\text{Confidence}(\{Milk, Bread\} \rightarrow \{Butter\}) = \frac{\text{Support}(\{Milk, Bread, Butter\})}{\text{Support}(\{Milk, Bread\})} = \frac{0.25}{0.5} =$$

Interpretation: There is a 50% chance that a customer who buys {Milk, Bread} will also buy {Butter}

Market Basket Analysis

Key Terminologies

Lift

1. Lift measures the strength of an association rule compared to random chance

$$Lift(X \rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)\, Support(Y)} = \frac{Confidence(X \rightarrow Y)}{Support(Y)} = \frac{P(X \cap Y)}{P(X)\, P(Y)} \quad (1)$$

2. Lift normalizes confidence by the overall frequency of Y, helping distinguish true association from coincidental co-occurrence

Market Basket Analysis

Key Terminologies

Lift

1. Lift helps assess whether the presence of one itemset positively, negatively, or independently affects the presence of another itemset

Table: Summary of Lift Values and Their Interpretation

Condition	Lift Value	Interpretation
$P(A \cap B) = 0$	0 (minimum)	A and B never occur together (mutually exclusive)
$P(A \cap B) = P(A)P(B)$	1	A and B are independent
$P(A \cap B) > P(A)P(B)$	> 1	Positive association (co-occur more than expected)
$P(A \cap B) < P(A)P(B)$	< 1	Negative association (co-occur less than expected)

Market Basket Analysis

Key Terminologies

Lift: Example

Consider the following transactions:

- Transaction 1: {Milk, Bread}
- Transaction 2: {Milk, Bread, Butter}
- Transaction 3: {Bread, Butter}
- Transaction 4: {Milk, Butter}

Step 1: Compute Support Values

- Support({Milk, Bread}) = 0.5
- Support({Butter}) = 0.75
- Support({Milk, Bread, Butter}) = 0.25

Market Basket Analysis

Key Terminologies

Lift: Example Step 2: Compute Lift

$$\text{Lift}(\{Milk, Bread\} \rightarrow \{Butter\}) = \frac{0.25}{0.5 \times 0.75} = \frac{0.25}{0.375} \approx 0.67$$

Interpretation: Since the lift is less than 1, this indicates a negative association — purchasing {Milk, Bread} reduces the likelihood of purchasing {Butter}

Market Basket Analysis

Frequent Item Generation

1. Foundational step in market basket analysis where combinations of items that appear together frequently are identified
2. **Frequent Itemset:** An itemset that meets or exceeds a predefined **minimum support threshold**

Market Basket Analysis

Frequent Item Generation

Steps

1. **Candidate Generation:**
   - Start by identifying individual items (1-itemsets) that meet the support threshold
   - Iteratively combine these items to generate larger itemsets (2-itemsets, 3-itemsets, etc.)
2. **Pruning:**
   - Eliminate itemsets that fail to meet the minimum support threshold; a superset cannot be more frequent than its subset
   - This reduces the number of candidate itemsets in subsequent iterations
3. **Final Frequent Itemsets:**
   - Only itemsets that satisfy the support condition are considered frequent

Market Basket Analysis

Frequent Item Generation

Steps

1. Transactions: {Milk, Bread}, {Milk, Bread, Butter}, {Bread, Butter}, {Milk, Butter}
2. Support Threshold = 50%
3. Frequent Itemsets:

$$\{Milk\}, \{Bread\}, \{Butter\}, \{Milk, Bread\}, \{Milk, Butter\}, \{Bread, Butter\}$$

Market Basket Analysis

Apriori Algorithm

1. Popular method for finding frequent itemsets in large transactional datasets and generating association rules
2. The Apriori principle is the foundation of frequent itemset generation; it is what makes the process computationally feasible
3. Uses the **Apriori Property** — All subsets of a frequent itemset must also be frequent; equivalently, if a subset is infrequent, all its supersets are infrequent
4. Applications include market basket analysis, healthcare and fraud detection

Market Basket Analysis

Apriori Algorithm

Apriori Property

1. All non-empty subsets of a frequent itemset must also be frequent
   - If {Milk, Bread, Butter} is frequent, then:

     $\{Milk, Bread\}, \{Milk, Butter\}, \{Bread, Butter\}$must also be frequent

2. Apriori principle is the pruning strategy; frequent itemset generation is the search process
3. Frequent itemset generation defines the procedure to explore itemsets, while the Apriori principle acts as a filtering rule that prunes unpromising candidates, making the search efficient

Market Basket Analysis

Apriori Algorithm

Apriori Property: Steps

1. **Step 1:** Identify all 1-itemsets that satisfy the minimum support threshold
2. **Step 2:** Generate 2-itemsets by combining frequent 1-itemsets
3. **Step 3:** Prune itemsets that fail to meet the minimum support threshold
4. **Step 4:** Repeat the process for larger itemsets (3-itemsets, 4-itemsets, etc.)
5. **Step 5:** Generate association rules from the frequent itemsets
6. **Step 6:** Evaluate the rules using confidence and lift metrics

Market Basket Analysis

Apriori Algorithm

Transaction Dataset

- T1: {Milk, Bread, Butter}
- T2: {Milk, Bread}
- T3: {Bread, Butter}
- T4: {Milk, Butter}
- T5: {Milk, Bread, Butter}

Minimum Support Threshold: 40%
Goal: Find frequent itemsets and generate association rules

Market Basket Analysis

Apriori Algorithm

Counting 1-itemsets Support

- $\{\text{Milk}\}$: $4/5 = 0.8$
- $\{\text{Bread}\}$: $4/5 = 0.8$
- $\{\text{Butter}\}$: $4/5 = 0.8$

Frequent 1-itemsets: $\{\text{Milk}\}$, $\{\text{Bread}\}$, $\{\text{Butter}\}$

Market Basket Analysis

Apriori Algorithm

Counting 2-itemsets Support:

- {Milk, Bread}: $3/5 = 0.6$
- {Milk, Butter}: $3/5 = 0.6$
- {Bread, Butter}: $3/5 = 0.6$

Frequent 2-itemsets: {Milk, Bread}, {Milk, Butter}, {Bread, Butter}

Market Basket Analysis

Apriori Algorithm

Counting 3-itemsets Support:

- {Milk, Bread, Butter}: $2/5 = 0.4$

Frequent 3-itemsets: {Milk, Bread, Butter}

Market Basket Analysis

Apriori Algorithm

Example Rules with Confidence:

- $\{\text{Milk, Bread}\} \rightarrow \{\text{Butter}\} : \text{Confidence} = \frac{2}{3} = 0.67$
- $\{\text{Milk, Butter}\} \rightarrow \{\text{Bread}\} : \text{Confidence} = \frac{2}{3} = 0.67$
- $\{\text{Bread, Butter}\} \rightarrow \{\text{Milk}\} : \text{Confidence} = \frac{2}{3} = 0.67$

Market Basket Analysis

Apriori Algorithm

Example Rules with Lift

$$\text{Lift}(\{Milk, Bread\} \rightarrow \{Butter\}) = \frac{\text{S}(\{Milk, Bread, Butter\})}{\text{S}(\{Milk, Bread\}) \times \text{S}(\{Butter\})}$$

$$\text{Lift}(\{Milk, Bread\} \rightarrow \{Butter\}) = \frac{0.4}{0.6 \times 0.8} \approx 0.83$$

Interpretation: Since Lift < 1, this is a weak association

Market Basket Analysis

Apriori Algorithm

Limitations

1. Computational Complexity
   - The algorithm performs multiple passes over the dataset, one for each level of itemset size, making it *computationally expensive* for large datasets.
   - As the number of items increases, the number of possible itemsets grows exponentially, resulting in a *combinatorial explosion*
2. High Memory Consumption
   - The algorithm generates a large number of *candidate itemsets*, many of which are later discarded
   - This leads to significant memory overhead, especially in datasets with a *large number of distinct items*

Market Basket Analysis

Apriori Algorithm

Limitations

1. Inefficiency with Low Support Thresholds
   - Setting a *low support threshold* increases the number of candidate itemsets drastically
   - This results in extensive computation and slower performance
2. Weak Performance on Sparse Data
   - In sparse datasets (e.g., online retail data), meaningful itemsets are often spread across many transactions
   - The Apriori algorithm may struggle to efficiently identify such itemsets

Introduction to Rule Generation

1. Rule generation is a critical step in **Market Basket Analysis**
2. It identifies actionable insights from frequent itemsets
3. Common metrics include **support**, **confidence**, and **lift**

Why Rule Generation?

1. Helps in identifying strong associations between products
2. Supports decision-making in marketing, sales, and inventory management
3. Drives personalized recommendations and targeted promotions

Rule Generation

Key Metrics for Rule Generation

1. **Support**: Fraction of transactions containing an itemset
2. **Confidence**: Probability that item B is purchased given item A is purchased
3. **Lift**: Measures the strength of a rule by comparing its occurrence with independence
4. **Leverage:** Measures how much more frequently X and Y occur together compared to what we would expect if they were independent
5. **Conviction:** measures how strongly the rule $X \rightarrow Y$ avoids making wrong predictions, compared to what would happen if X and Y were independent; measures how reliable a rule is by looking at how rarely it fails

Rule Generation

Key Metrics for Rule Generation

Leverage

1. Measures how much more frequently X and Y occur together compared to what we would expect if they were independent

$$\text{Leverage}(X \rightarrow Y) = \text{Support}(X \cup Y) - (\text{Support}(X) \times \text{Support}(Y))$$

- 0: Independence between X and Y
- Positive: Positive association
- Negative: Negative association

Rule Generation

Key Metrics for Rule Generation

1. Given a dataset with 1000 transactions
   - $Support(X) = 0.3$
   - $Support(Y) = 0.4$
   - $Support(X \cup Y) = 0.2$
2. Compute Leverage:

$$\begin{aligned} \text{Leverage}(X \rightarrow Y) &= \text{Support}(X \cup Y) - (\text{Support}(X) \times \text{Support}(Y)) \\ &= 0.2 - (0.3 \times 0.4) \\ &= 0.2 - 0.12 = 0.08 \end{aligned}$$

Rule Generation

Key Metrics for Rule Generation

1. **Conviction:** Measures the dependency between X and Y by comparing expected and actual frequencies of incorrect predictions

$$\text{Conviction}(X \rightarrow Y) = \frac{1 - \text{Support}(Y)}{1 - \text{Confidence}(X \rightarrow Y)}$$

2. Numerator: How often Y does NOT occur in general; baseline probability of failure
3. Denominator: Probability that Y does not appear given X; observed failure rate of the rule
4. Higher conviction values indicate that the rule fails less often than expected by chance

Rule Generation

Key Metrics for Rule Generation

1. Given a dataset with $Support(X) = 0.4$, $Support(Y) = 0.5$, $Support(X \cup Y) = 0.3$
2. Compute Conviction:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)} = \frac{0.3}{0.4} = 0.75$$

$$\text{Conviction}(X \rightarrow Y) = \frac{1 - \text{Support}(Y)}{1 - \text{Confidence}(X \rightarrow Y)}$$

$$= \frac{1 - 0.5}{1 - 0.75} = \frac{0.5}{0.25} = 2$$

Rule Generation

Key Metrics for Rule Generation

Conviction

Table: Interpreting conviction Values

Conviction Value	Condition	Interpretation
0	$Support(Y) = 1$	Y always occurs; rule is not informative
< 1	$Confidence(X \rightarrow Y) <$ Support(Y)	Negative association; rule fails more than expected
$= 1$	$Confidence(X \rightarrow Y) = Support(Y)$	No association; X and Y are independent
> 1	$Confidence(X \rightarrow Y) > Support(Y)$	Positive association; rule is reliable
$\rightarrow \infty$	$Confidence(X \rightarrow Y) = 1$	Perfect rule; no failures

Rule Generation

Establishing an Association

1. A strong association rule should have:
   - High **Support** - The rule applies to many transactions
   - High **Confidence** - The rule is frequently correct
   - High **Lift** - The rule is significant beyond random chance
   - Positive **Leverage** - The co-occurrence is greater than expected
   - High **Conviction** - The rule rarely fails